

JOB APPLICATION TRACKER

1. Project Overview

The **Job Application Tracker** is a console-based Python application developed using Object-Oriented Programming (OOP) and modular programming concepts. The application allows users to manage profiles, job listings, and job applications while validating eligibility based on GPA and required skills. It uses JSON files for data storage, custom exceptions for validation, and recursive job recommendations based on partial skill matching.

2. Problem Statement

Develop a modular Python application that manages user profiles, job listings, and job applications. The application should support CRUD operations, validate job eligibility based on GPA and required skills, store data using JSON files, handle errors using custom exceptions, and recommend suitable jobs using a recursive function based on partial skill matching.

3. Features

User Profile Management

- Create user profiles
- View user profiles
- Update user profiles
- Delete user profiles

Job Management

- Add job listings
- View available jobs
- Update job details
- Delete job listings

Application Processing

- Apply for jobs

- Validate minimum GPA
- Validate required skills
- Store successful applications with date and time
- View submitted applications

Job Recommendations

- Recommend jobs based on partial skill matching
- Use recursion for job recommendations

Data & Error Handling

- Store data using JSON files
- Handle custom exceptions
- Modular code organization

4. Technologies Used

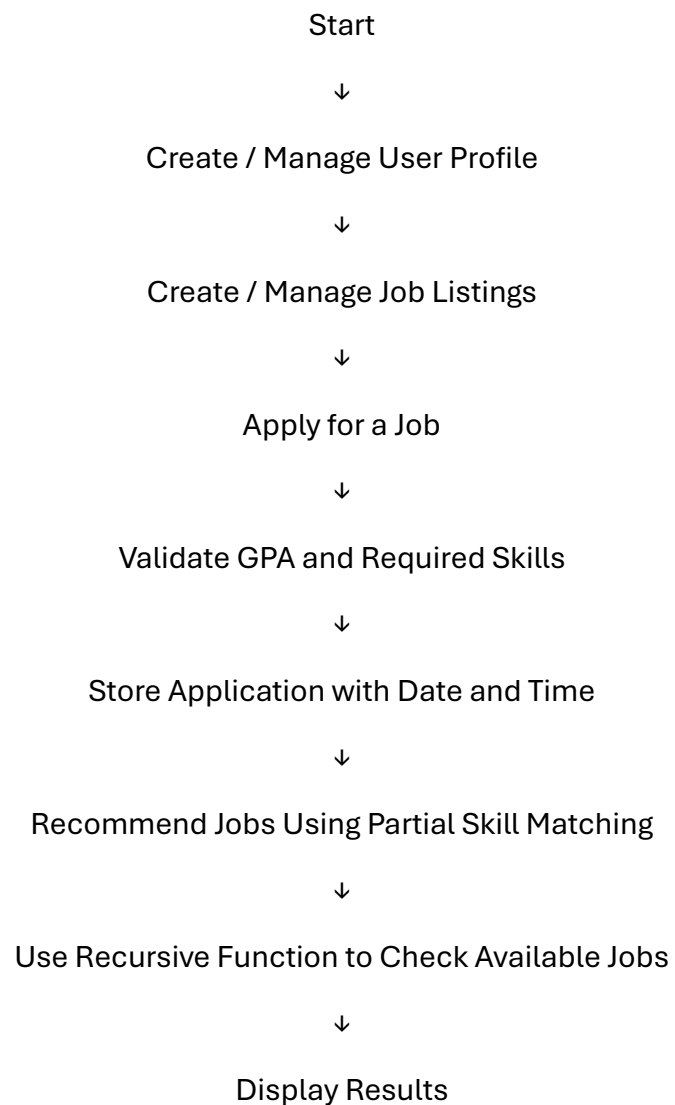
- Python 3
- Visual Studio Code
- JSON
- Object-Oriented Programming
- Modular Programming
- Command Prompt / Terminal

5. Python Concepts Used

- Object-Oriented Programming (OOP)
- Classes and Objects
- Constructors (`__init__`)
- Modular Programming
- JSON File Handling
- File Handling
- Custom Exceptions
- Exception Handling

- CRUD Operations
- Lists
- Dictionaries
- Loops
- Conditional Statements
- User Input Handling
- Datetime Module
- Menu-Driven Programming
- Recursion

6. Program Workflow

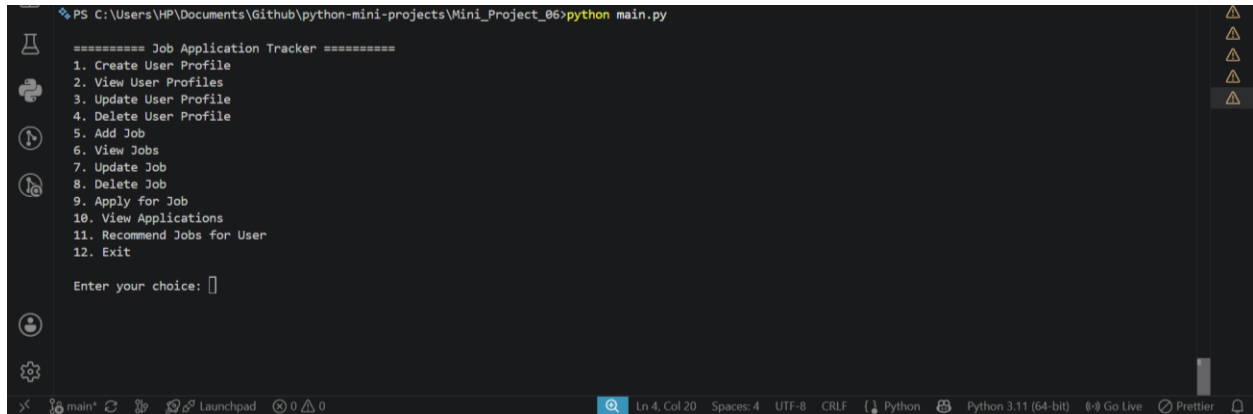


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End

7. Output Screenshots and Explanation

Screenshot 1: Main Menu



```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06>python main.py

===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

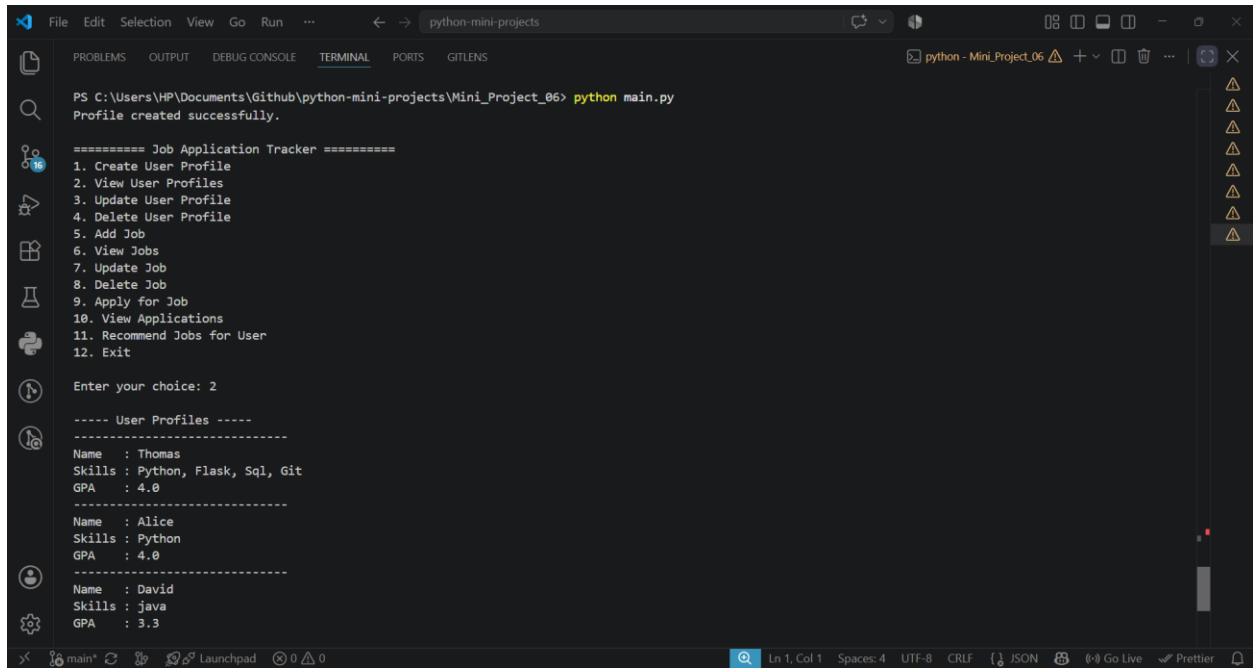
Enter your choice: 
```

The screenshot shows a terminal window with a dark background. The title bar indicates the file path is C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06 and the command executed is python main.py. The output displays a menu for the 'Job Application Tracker' with 12 numbered options. The options are: 1. Create User Profile, 2. View User Profiles, 3. Update User Profile, 4. Delete User Profile, 5. Add Job, 6. View Jobs, 7. Update Job, 8. Delete Job, 9. Apply for Job, 10. View Applications, 11. Recommend Jobs for User, and 12. Exit. Below the menu, there is a prompt 'Enter your choice:' followed by a cursor. The terminal window has a sidebar on the left with various icons and a status bar at the bottom showing 'Ln 4, Col 20', 'Spaces: 4', 'UTF-8', 'CRLF', 'Python', 'Python 3.11 (64-bit)', 'Go Live', 'Prettier', and a bell icon.

Explanation

The application starts by displaying a menu-driven interface with options to manage user profiles, job listings, and job applications. Users can choose different operations by entering the corresponding menu number.

Screenshot 2: User Profile Management



The screenshot shows a VS Code terminal window with the following output:

```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06> python main.py
Profile created successfully.

===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

Enter your choice: 2

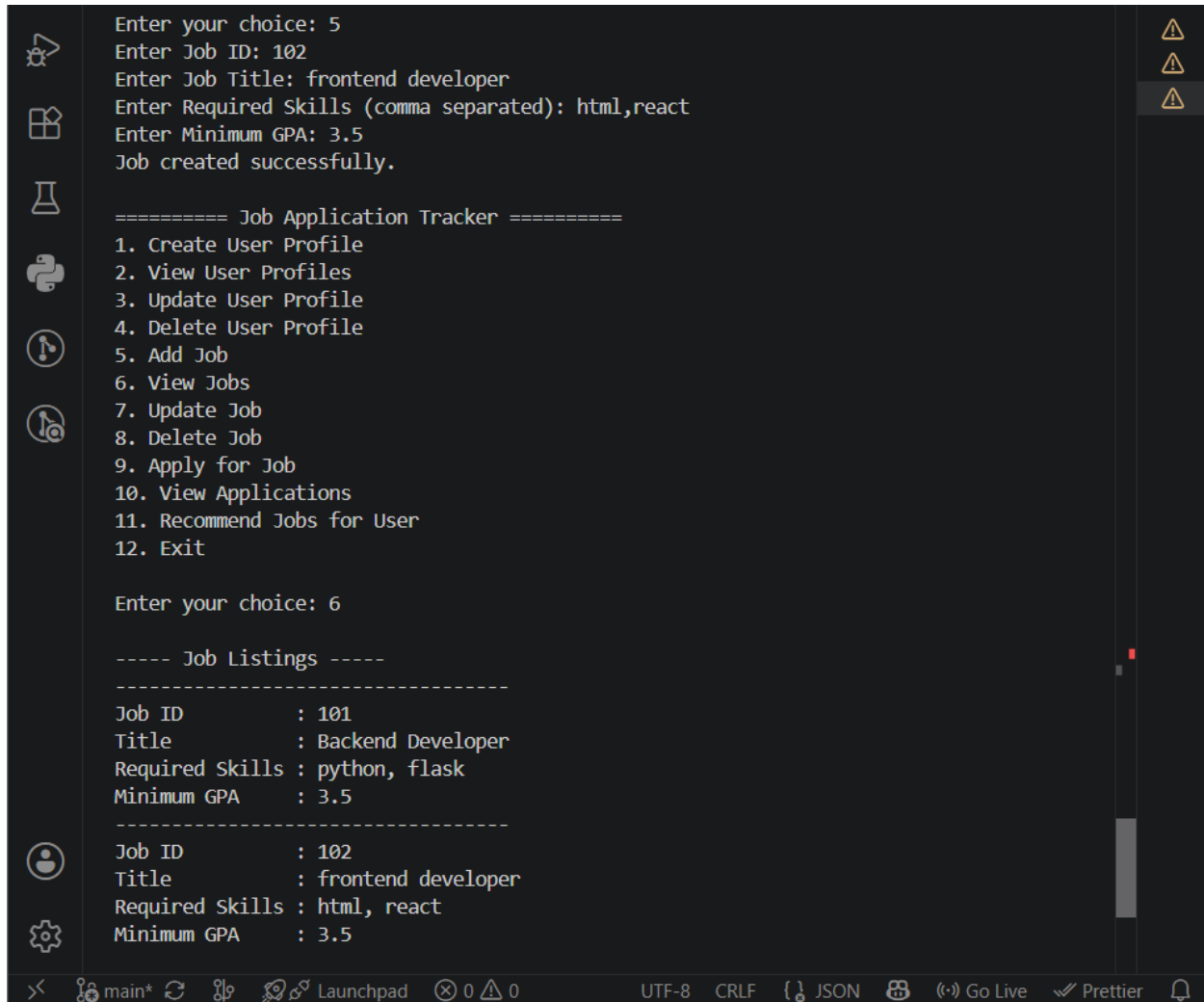
----- User Profiles -----
Name   : Thomas
Skills : Python, Flask, Sql, Git
GPA    : 4.0
-----
Name   : Alice
Skills : Python
GPA    : 4.0
-----
Name   : David
Skills : java
GPA    : 3.3
```

The terminal window is titled "python-mini-projects" and shows the execution of a Python script. The output displays a menu of options for the "Job Application Tracker" application. The user has selected option 2, "View User Profiles", which has resulted in a list of three user profiles: Thomas, Alice, and David. Each profile includes their name, skills, and GPA.

Explanation

The application allows users to create, view, update, and delete user profiles. Each profile stores the user's name, skills, and GPA, which are later used to validate eligibility during the job application process.

Screenshot 3: Job Management



```
Enter your choice: 5
Enter Job ID: 102
Enter Job Title: frontend developer
Enter Required Skills (comma separated): html,react
Enter Minimum GPA: 3.5
Job created successfully.

===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

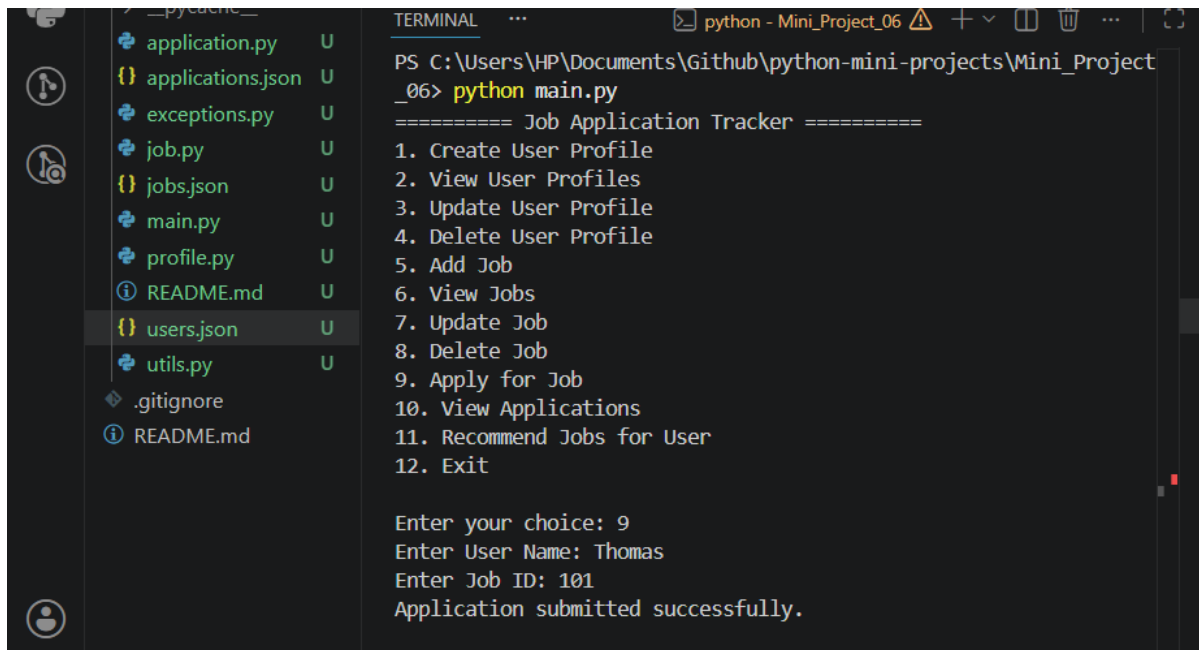
Enter your choice: 6

----- Job Listings -----
-----
Job ID      : 101
Title       : Backend Developer
Required Skills : python, flask
Minimum GPA  : 3.5
-----
Job ID      : 102
Title       : frontend developer
Required Skills : html, react
Minimum GPA  : 3.5
```

Explanation

The application provides functionality to add, view, update, and delete job listings. Each job record contains a Job ID, job title, required skills, and the minimum GPA required for eligibility.

Screenshot 4: Successful Job Application



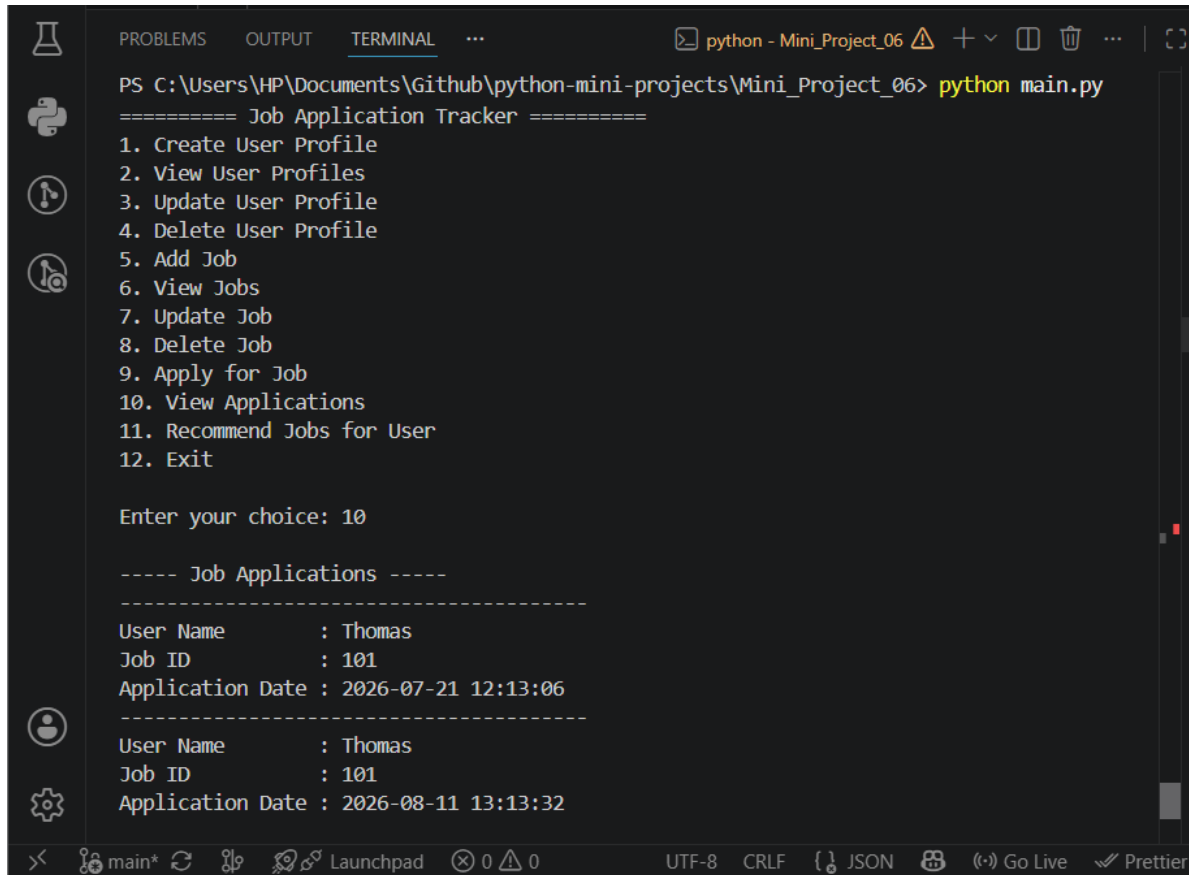
```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06> python main.py
===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

Enter your choice: 9
Enter User Name: Thomas
Enter Job ID: 101
Application submitted successfully.
```

Explanation

After validating the user's GPA and required skills, the application successfully submits the job application and records the application date and time. The submitted application is stored in the applications.json file.

Screenshot 5: Viewing Applications



```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06> python main.py
===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

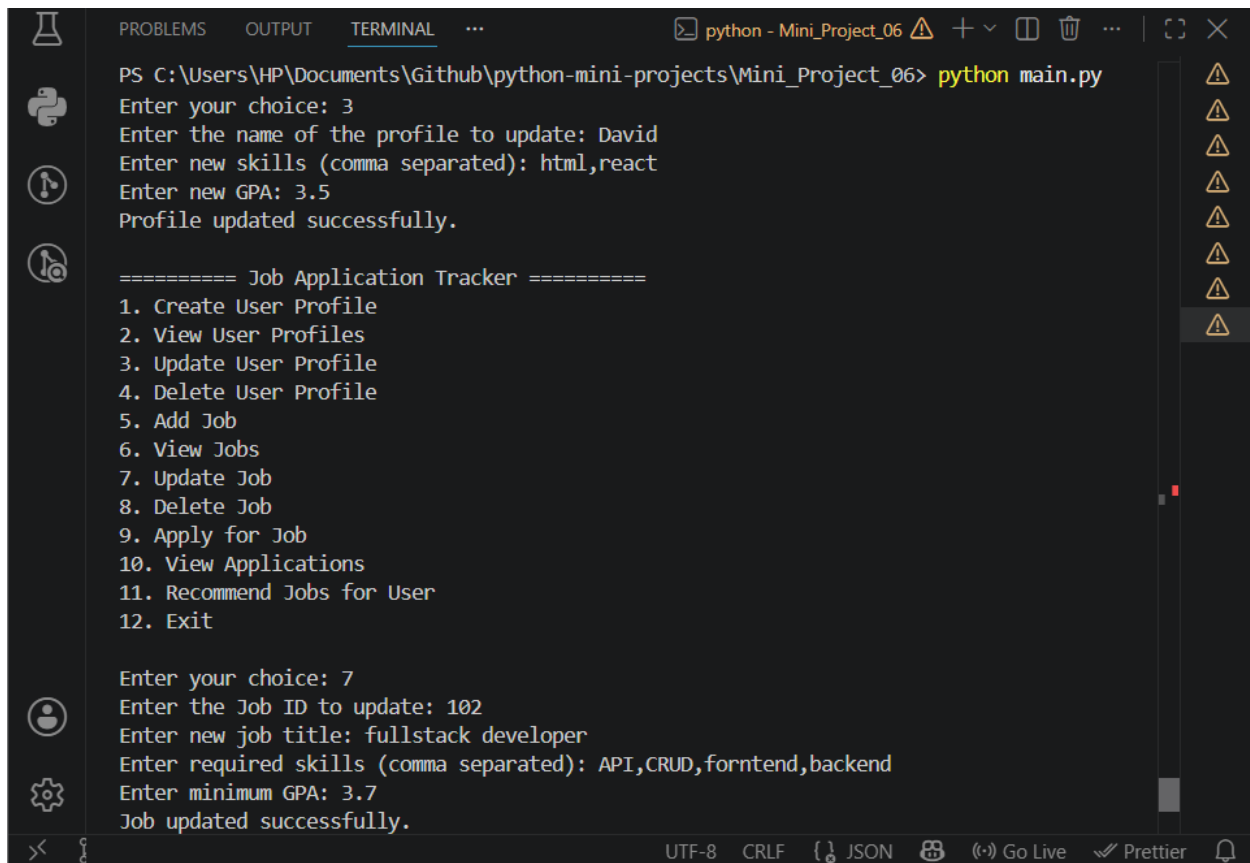
Enter your choice: 10

----- Job Applications -----
-----
User Name      : Thomas
Job ID        : 101
Application Date : 2026-07-21 12:13:06
-----
User Name      : Thomas
Job ID        : 101
Application Date : 2026-08-11 13:13:32
```

Explanation

The application displays all submitted job applications along with the applicant's name, Job ID, and application date. This allows users to review previously submitted applications.

Screenshot 6: Updating User and Job Information



The screenshot shows a VS Code terminal window with the following content:

```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06> python main.py
Enter your choice: 3
Enter the name of the profile to update: David
Enter new skills (comma separated): html,react
Enter new GPA: 3.5
Profile updated successfully.

===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

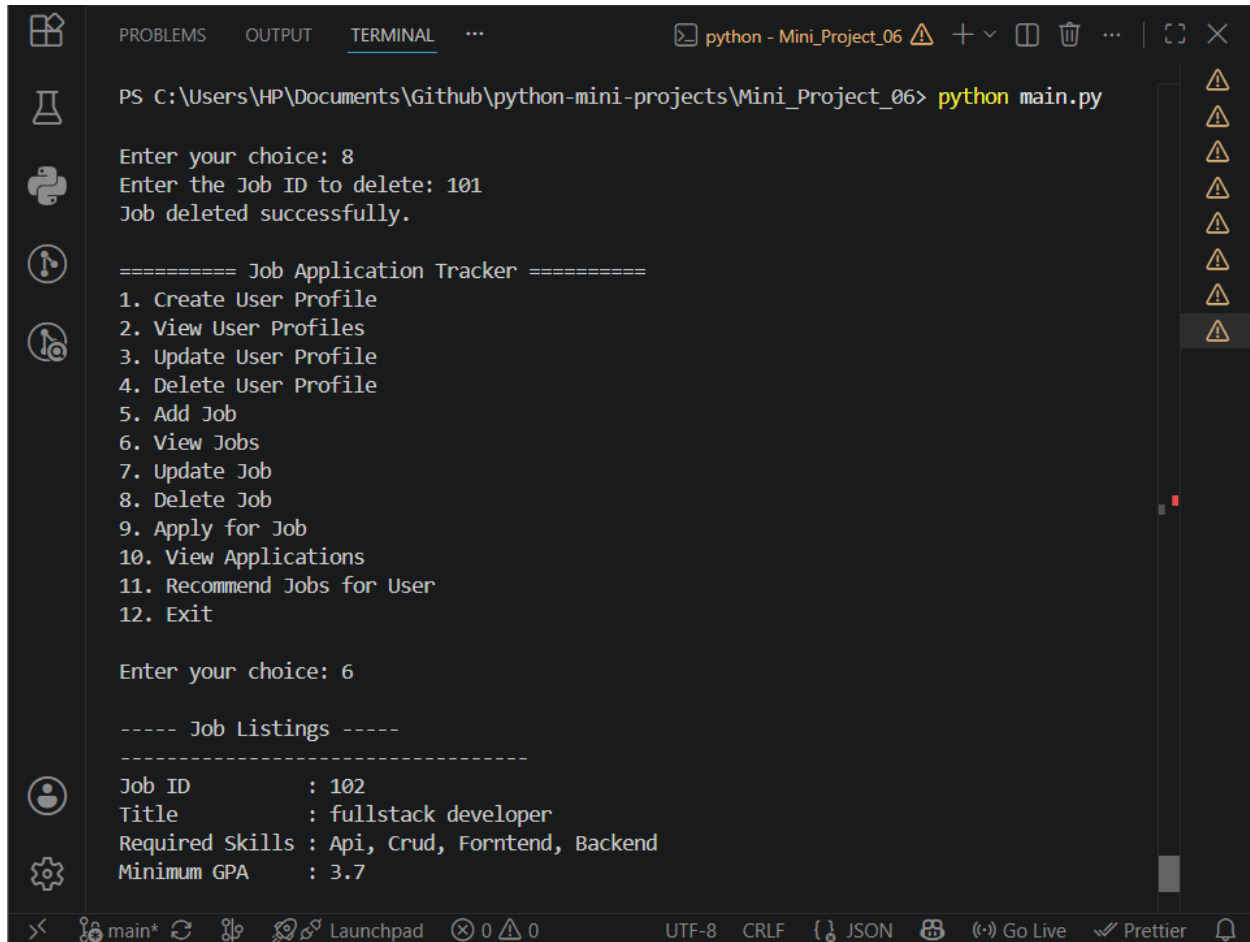
Enter your choice: 7
Enter the Job ID to update: 102
Enter new job title: fullstack developer
Enter required skills (comma separated): API,CRUD,forntend,backend
Enter minimum GPA: 3.7
Job updated successfully.
```

The terminal window has a dark theme and includes a sidebar on the left with icons for Explorer, Search, and Run and Debug. The top bar shows the file name 'python - Mini_Project_06' and various window controls. The bottom status bar indicates the encoding is UTF-8, line endings are CRLF, and the file is a JSON document. It also shows icons for Go Live and Prettier.

Explanation

The application allows users to modify existing profile details and job information. The updated data is saved to the corresponding JSON files, ensuring that future operations use the latest information.

Screenshot 7: Deleting User and Job Records



The screenshot shows a VS Code terminal window with the following content:

```
PS C:\Users\HP\Documents\Github\python-mini-projects\Mini_Project_06> python main.py

Enter your choice: 8
Enter the Job ID to delete: 101
Job deleted successfully.

===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

Enter your choice: 6

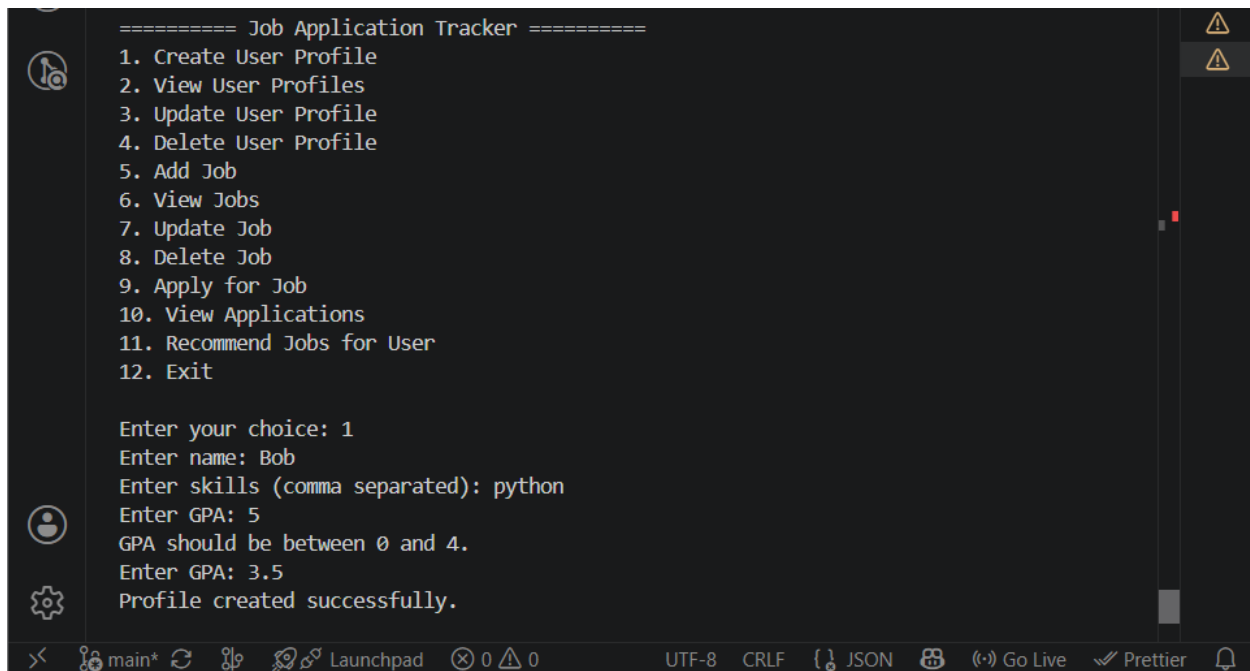
----- Job Listings -----
-----
Job ID      : 102
Title       : fullstack developer
Required Skills : Api, Crud, Forntend, Backend
Minimum GPA  : 3.7
```

The terminal window has a dark theme and shows the execution of a Python script. The user has entered choice 8 to delete a job and choice 6 to view job listings. The output shows that job ID 101 was successfully deleted and the current job listings include job ID 102.

Explanation

The application provides options to remove user profiles and job listings. After deletion, the corresponding records are permanently removed from the JSON files, while previously submitted application records remain unchanged.

Screenshot 8: Exception Handling



```
===== Job Application Tracker =====
1. Create User Profile
2. View User Profiles
3. Update User Profile
4. Delete User Profile
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

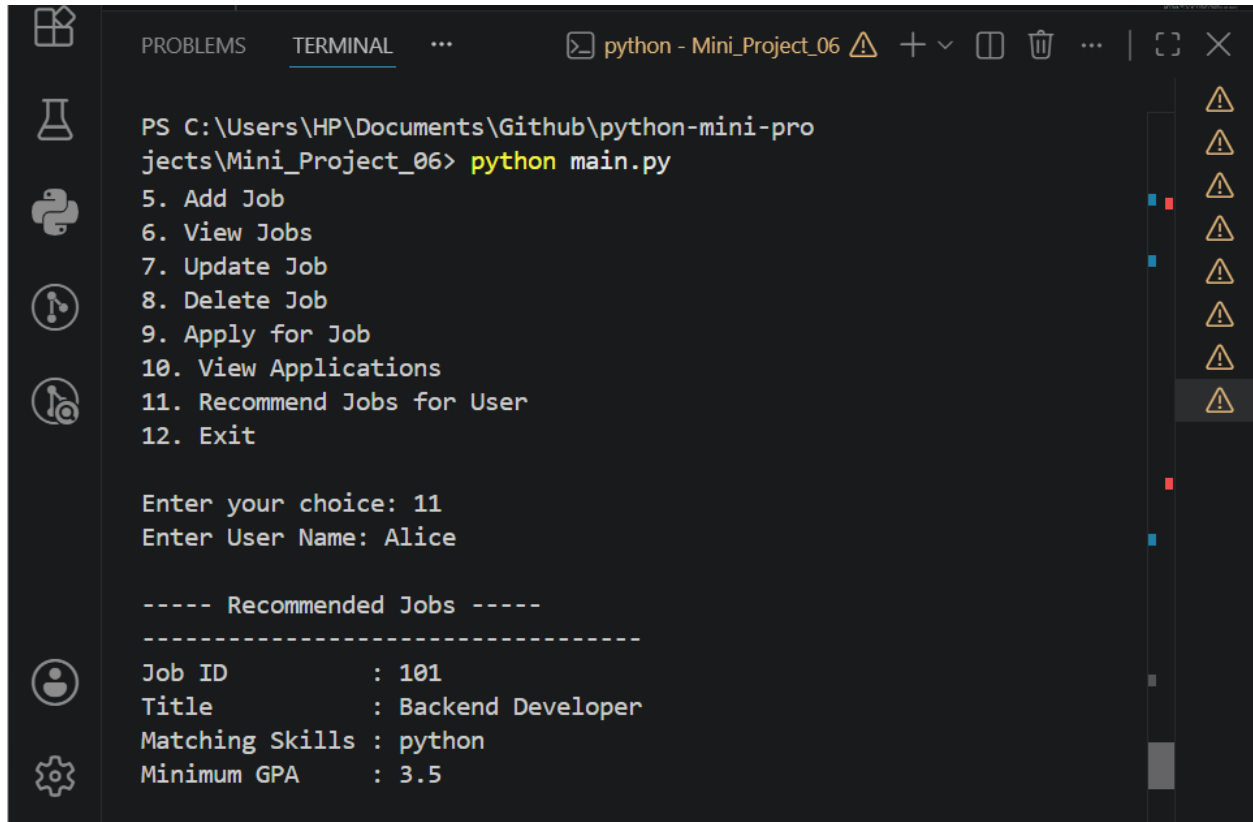
Enter your choice: 1
Enter name: Bob
Enter skills (comma separated): python
Enter GPA: 5
GPA should be between 0 and 4.
Enter GPA: 3.5
Profile created successfully.
```

The screenshot shows a code editor with a dark theme. The code is a Python script for a 'Job Application Tracker'. It features a menu with 12 options. The user has selected option 1, 'Create User Profile'. The program prompts for a name ('Bob'), skills ('python'), and GPA. The user entered '5' for GPA, which triggered an exception (GPA should be between 0 and 4). The user then entered '3.5', and the program successfully created the profile. The editor's status bar at the bottom shows 'main*', 'Launchpad', 'UTF-8', 'CRLF', 'JSON', 'Go Live', and 'Prettier'.

Explanation

The application validates user input and handles exceptions gracefully. It displays appropriate messages for scenarios such as duplicate user profiles, duplicate job IDs, insufficient GPA, missing required skills, invalid menu choices, or attempting to apply for non-existent jobs or user profiles. These validations ensure reliable and error-free execution.

Screenshot 9: Job Recommendations



```
PS C:\Users\HP\Documents\Github\python-mini-pro
jects\Mini_Project_06> python main.py
5. Add Job
6. View Jobs
7. Update Job
8. Delete Job
9. Apply for Job
10. View Applications
11. Recommend Jobs for User
12. Exit

Enter your choice: 11
Enter User Name: Alice

----- Recommended Jobs -----
-----
Job ID       : 101
Title        : Backend Developer
Matching Skills : python
Minimum GPA  : 3.5
```

Explanation:

The application recommends available jobs based on partial skill matching. A recursive function checks the user's skills against the required skills of each job and displays suitable job recommendations.

8. Learning Outcomes

After completing this project, I learned how to:

- Develop modular Python applications using multiple modules
- Apply Object-Oriented Programming concepts
- Implement CRUD operations using JSON files
- Create and use custom exceptions
- Validate user eligibility using GPA and skills
- Work with date and time using the datetime module

- Implement recursive functions for job recommendations
- Perform partial skill matching
- Build menu-driven console applications
- Organize code into reusable and maintainable modules

9. Conclusion

The Job Application Tracker successfully demonstrates the implementation of Object-Oriented Programming concepts, modular programming, JSON file handling, and custom exception handling in Python. The project provides an efficient way to manage user profiles, job listings, and job applications while ensuring proper validation of eligibility criteria. It also strengthens the understanding of CRUD operations, file handling, exception handling, and menu-driven application development. Overall, this project offers practical experience in designing modular, maintainable, and user-friendly Python applications.